

**EEU44C04 / CS4031 / CS7NS3 / EEP55C27**  
**Next Generation Networks**

**Wireless channel  
impairments and  
mitigation techniques**

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# Impairments and mitigation

## IMPAIRMENTS

干扰/损耗

- Path loss 路径损耗
- Shadowing 阴影效应
- Fading 衰落
- Multipath 多径效应
- Noise 噪声
- Interference 干扰

## MITIGATION

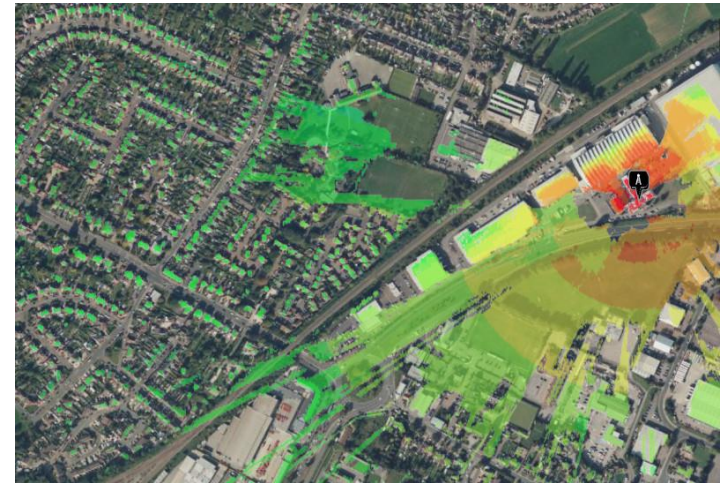
应对

- Diversity
- Directional antennas
- Coding and modulation
- Spread spectrum
- System-level mitigation: frequency and code planning, multiple access methods, etc.

# Path loss

路径损耗 (Path loss) : 信号传得越远就越弱, 哪怕中间没有任何障碍物挡着 (有视距/line-of-sight也一样) 。大致跟距离的平方成反比, 离得越远衰减越厉害。

- Even when line-of-sight exists, signal attenuates with **distance**
- **Path loss**: attenuation undergone by an electromagnetic wave in transit between a transmitter and a receiver
- Roughly proportional to  $1/d^2$ , where  $d$  is the TX-RX distance

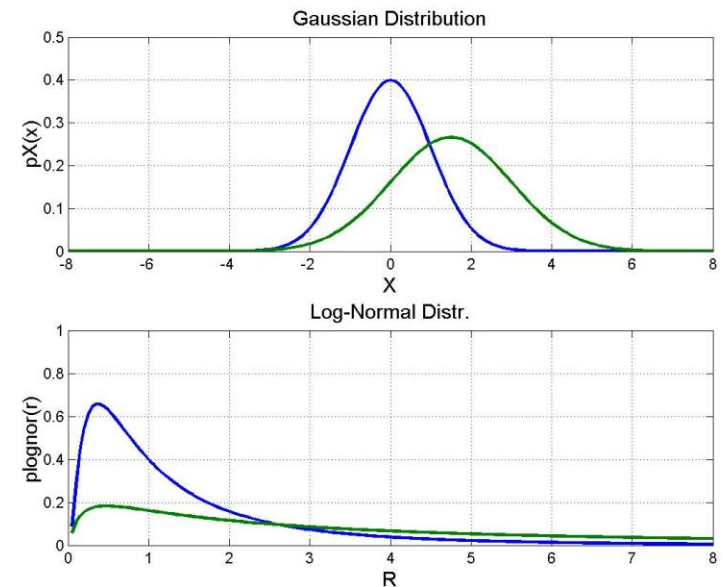


Source: <https://cloudrf.com/>

# Shadowing

阴影效应 (Shadowing)：信号被建筑物、树木这类障碍物挡住导致的随机衰减，因为障碍物的种类和数量是随机的，所以这个效应也是随机的，一般用对数正态分布来建模。

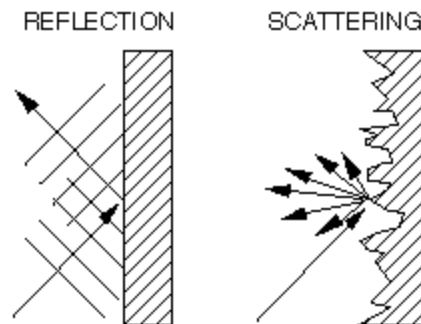
- Models attenuation from **obstacles**
- A random event, due to the random number and types of obstacles
- Typically modeled with a log-normal distribution
  - ✓ Local mean power, expressed in dB has a Normal (Gaussian) distribution



# Multipath

多径效应 (Multipath)：信号经过反射、散射后，会有好几份"副本"从不同路径到达接收端，到达时间还不一样（这叫delay spread），严重时不同副本会互相抵消。这个效应主要体现在频域上信号会变得不稳定。

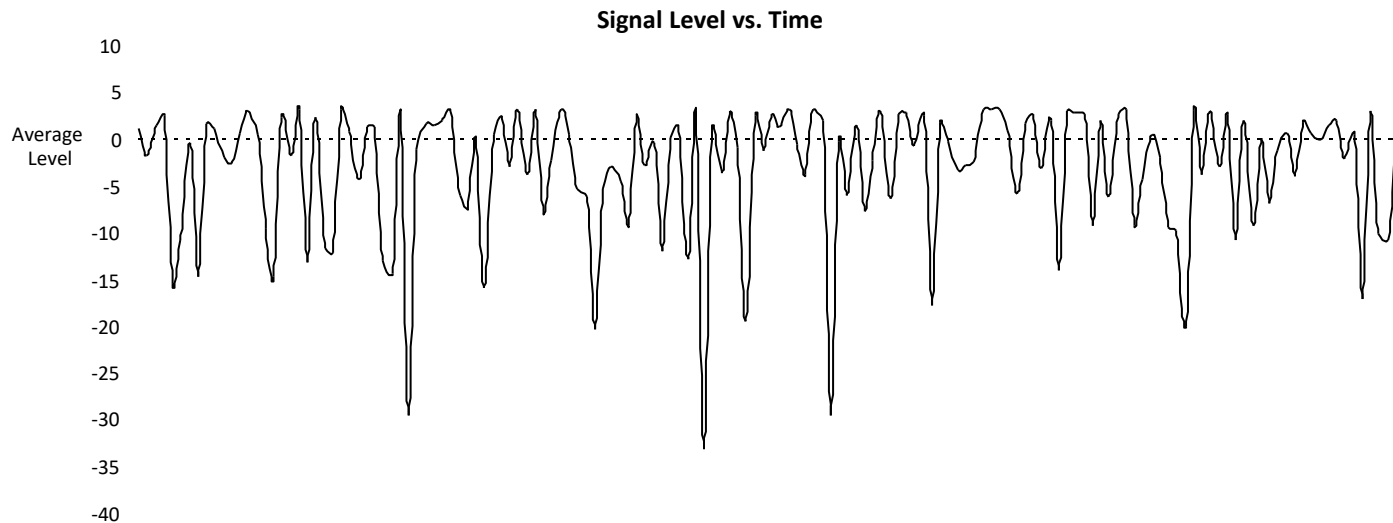
- Due to **reflections and scattering**, multiple versions of the transmitted signal may arrive at the receiver
  - ✓ In severe cases, out-of-phase versions may cancel each other
- **Delay spread**: original signal is spread due to different delays of parts of the signal
- Leads to variability in **frequency** domain



# Fading

衰落 (Fading)：由于收发双方的移动（比如你拿着手机走动），信道特性随时间快速变化，导致接收功率忽高忽低。这个是在时域上的不稳定，跟多径是“同一件事的两个角度”（时域 vs 频域）。

- Quick variations in received power due to time-varying channel characteristics (related to TX/RX **mobility**)
- Various stochastic models (Rayleigh, Rician, etc.) for indoor and outdoor environments
- Leads to variability in **time** domain



# Noise

噪声 (Noise) : 各种叠加在信号上的"干扰杂音", 来源可以是自然的 (闪电) 也可以是人为设备。  
常建模成白噪声 (各个频率上功率均匀分布)。用信噪比 (SNR) 衡量信道质量好坏

- Unwanted signals added to the message signal
- May be due to signals generated by **natural phenomena** e.g., lightning, **or man-made sources** e.g., transmitting and receiving equipment, as well as sparks in passing cars, wiring in thermostats, etc.
- Often modeled in the aggregate as a random signal in which power is distributed uniformly across all frequencies (white noise)
- Signal-to-Noise Ratio (SNR) often used as a metric in the assessment of channel quality

# Interference

干扰 (Interference) : 来自其他信号源的冲突, 比如同一个cell里别的手机、相邻cell的通话、或者用了同一频段的其他系统。  
分两种: 同频干扰 (CCI) 和邻频干扰 (ACI) 。

- **Sources** of interference:
  - Another mobile in the same cell
  - A call in progress in the neighbouring cell
  - Other base stations operating in the same frequency band
  - Non-cellular system leaks energy into the cellular frequency band
- **Two major kinds** of cellular interference:
  - Co-channel interference (CCI): caused by signals at the same frequency
  - Adjacent channel interference (ACI): caused by signals from neighbouring frequencies



# Impairments and mitigation

## IMPAIRMENTS

- Path loss
- Shadowing
- Fading
- Multipath
- Noise
- Interference

## MITIGATION

- Diversity
- Directional antennas
- Coding and modulation
- Spread spectrum
- System-level mitigation: frequency and code planning, multiple access methods, etc.

# Diversity

分集技术 (Diversity) : 核心思路是"多收几份独立的信号副本, 然后组合起来", 提高可靠性。  
可以是空间上的 (多天线)、频率上的 (多个频段)、时间上的 (重复发送) 或极化方式上的。

- Combine **independently received** versions of the desired signal
- Spatial diversity: multiple antennas
- Frequency diversity: multiple frequencies
- Temporal diversity: repeated transmissions
- Polarization diversity: different polarizations



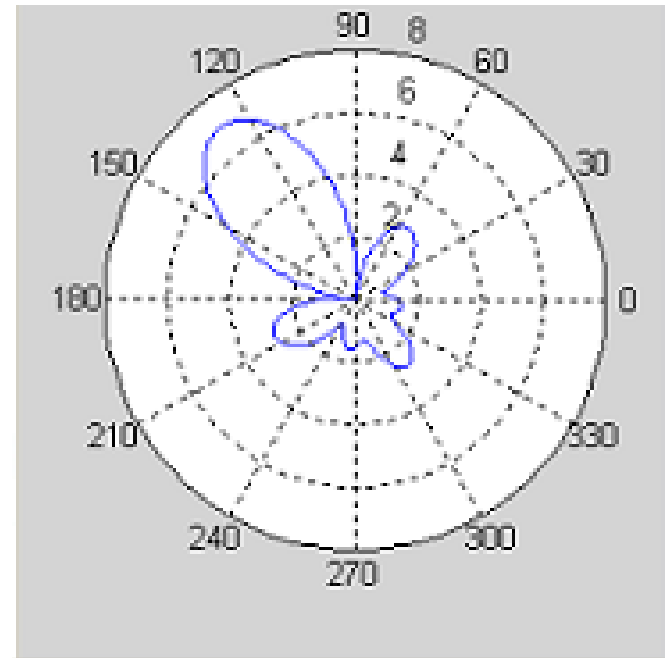
In spatial diversity, antennas are separated by at least half a wavelength ( $\lambda/2$ )

- Note:  $\lambda = c / f$ , where  $f$  is the frequency of the electromagnetic waves and  $c = 3 \times 10^8$  m/s

# Directional antennas

定向天线 (Directional antennas) : 把天线阵列排成特定形状, 让信号增益集中在某个方向、其他方向弱一些, 这样能减少干扰、也更难被截获或干扰。

- Antenna array elements arranged in a certain geometry
- By appropriately weighing the elements, one can **maximize the gain in one direction** while minimizing it in other directions
- Can be used to reduce interference (as well as probability of interception or jamming)



Example of antenna pattern produced by a 7-element, uniform (equally) spaced, circular array

# Coding and modulation

编码与调制 (Coding and modulation) : 调制方式在功率效率、频谱效率、抗干扰能力上各有取舍 (比如QPSK比16-QAM更省功率) 。多载波调制把一条高速数据流拆成多条低速流分别传, 这样频率选择性衰落只影响部分子载波而不是全部。另外, 前向纠错编码通过加"冗余"信息来抗信道错误——但冗余越多意味着编码率越低 (比如1/2意味着一半是纠错信息) 。

e.g., to achieve same Bit Error Rate (BER), QPSK uses less power than 16-QAM (lower SNR needed to get same error performance)

- Digital modulation schemes vary in...
  - ✓ Power efficiency – how much power is needed to transmit
  - ✓ **Spectral efficiency** – bits/second/Hz
  - ✓ Robustness – to multipath fading, noise, interference
- Multi-carrier modulation splits the bit stream into several lower bit rate streams, each sent using an **independent carrier frequency**
  - ✓ Frequency selective fading only affects some subcarriers, not the entire signal
- Forward error correction mitigates the effect of channel errors

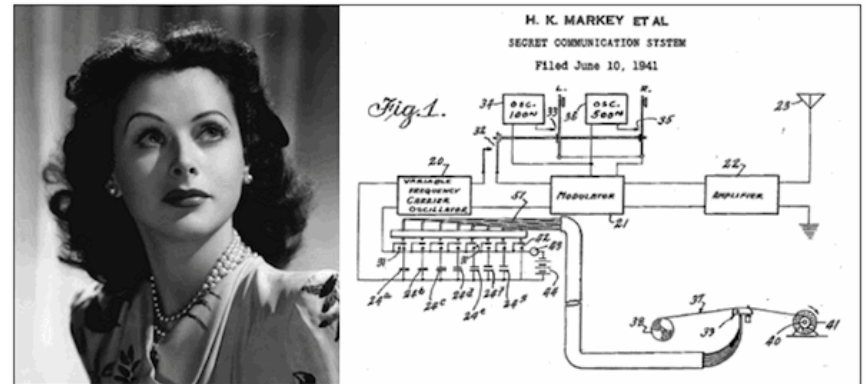
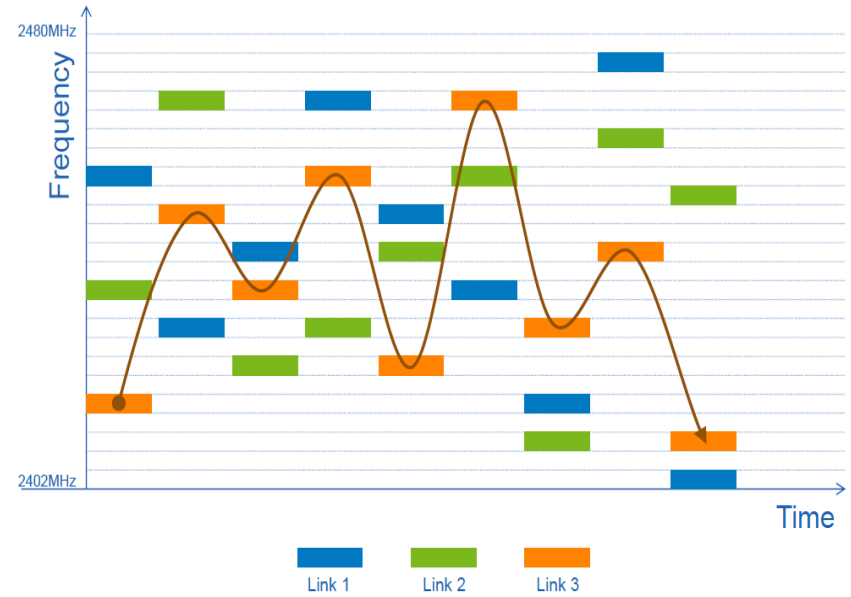
# Spread spectrum

扩频 (Spread spectrum) : 把信号"摊开"到很宽的频率范围传输, 接收端再收集回来, 因为摊开后信号很像噪声, 很难被检测或干扰。最早是军用 (抗干扰、抗截获), 后来民用也普及了。分两种: 跳频 (Frequency hopping, 收发双方按约定序列在不同频段间跳) 和直接序列扩频 (DSSS, 用户比特流跟一个"chipping sequence"做异或)。DSSS实现更复杂但抗多径衰落能力更强, 也更难被侦测。

- Spread spectrum signals are distributed over a **wide range of frequencies** and then collected back at the receiver
  - ✓ These wideband signals are noise-like, hence difficult to detect or interfere with
- Initially adopted in military applications, for its resistance to jamming and difficulty of interception
- More recently, adopted in commercial wireless communications
- Frequency hopping and Direct sequence spread spectrum variants

# Frequency hopping

- Transmitter and receiver **hop among different frequency bands**, according to a pre-established and agreed hopping sequence
- Slow hopping (one frequency for several bit periods), fast hopping (hopping during a bit period)



# Direct sequence spread spectrum (1)

- User bit stream is XORed with a chipping sequence

The diagram illustrates the relationship between the spreading factor  $S$  and the timing parameters of the spread spectrum system. A box labeled "Spreading factor" has a blue arrow pointing to the numerator  $t_b$  of the equation  $S = \frac{t_b}{t_c}$ . Another box labeled "Bit period" has a blue arrow pointing to  $t_b$ . A third box labeled "Chip period" has a blue arrow pointing to the denominator  $t_c$ . To the right of the equation, the inequality  $t_c \ll t_b$  is shown.

$$S = \frac{t_b}{t_c}$$

$t_c \ll t_b$

Spreading factors vary from 4 to 512 in FDD UMTS (3G), up to 4,096 in LoRa systems (IoT)

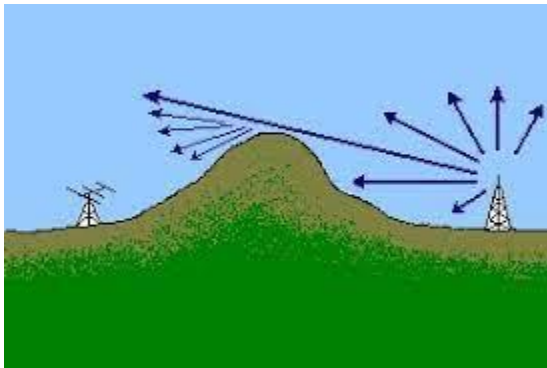
## Direct sequence spread spectrum (2)

- The **spreading factor** determines the bandwidth of the signal to be transmitted
  - ✓ Up to order of 1,000 for commercial applications
  - ✓ Up to 10,000 for military applications
- Example: IEEE 802.11 (Barker code)
- Frequency Hopping Spread Spectrum (FHSS) is simpler to implement than Direct Sequence Spread Spectrum (DSSS)
- DSSS is more resistant to fading and multipath and much harder to detect



不同环境下最麻烦的impairment不一样：LAN里主要是干扰（接入点太密集），WAN里主要是路径损耗和噪声（基站间隔远、覆盖面积大），丘陵地区主要是阴影（山挡住了信号），室内主要是多径（反射物太多）。

- What is the most detrimental wireless channel impairment in Local Area Networks (LAN)?
- And in Wide Area Networks (WAN)?
- How about a hilly region?
- And for indoor environments?



What is the most detrimental wireless channel impairment in LAN? → **interference (access points are deployed close to each other)**

And in WAN? → **path loss, noise (base stations are deployed sufficiently far apart, but coverage area is large)**

How about a hilly region? → **shadowing "hiding" the transmitter (from the receiver's viewpoint)**

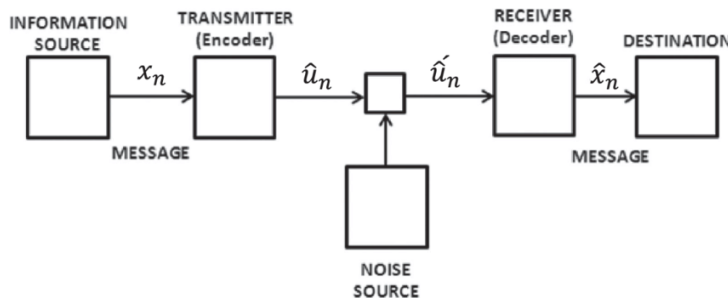
And for indoor environments? → **multipath due to many reflections of objects present in indoor environments**

Which of these coding rates is least beneficial in case of unreliable channel conditions?

☐ 1

☐ 1/3

☐ 1/2



Which of these coding rates is least beneficial in case of unreliable channel conditions?

✓ 1

☐ 1/3

☐ 1/2

**Code rate = 1 means we add no redundancy to the information to be transmitted, which instead is needed to protect the information if the channel is unreliable, i.e., likely to introduce errors.**

Explain the meaning of the following equation

$$C = B \log_2 \left( 1 + \frac{S}{I + N} \right)$$



- Is it always beneficial to increase  $B$ ? (consider that the noise power is  $N = N_0 B$ , where  $N_0$  is the noise's power spectral density). Motivate your answer.
- Is it always beneficial to increase the transmit power? (keep in mind the broadcast and shared nature of wireless medium). Motivate your answer.

Explain the meaning of the following equation

$$C = B \log_2 \left( 1 + \frac{S}{I + N} \right)$$

$C$  = channel capacity [bps]

$B$  = system bandwidth [Hz]

$S$  = useful signal power [W]

$I$  = interfering signal power [W]

$N$  = noise power [W]

- Is it always beneficial to increase  $B$ ? (consider that the noise power is  $N=N_0B$ , where  $N_0$  is the noise's power spectral density). Motivate your answer. (there is a trade-off between increasing the bandwidth (more data rate potentially achievable but also more noise overall) and decreasing it (the converse))
- Is it always beneficial to increase the transmit power? (keep in mind the broadcast and shared nature of wireless medium). Motivate your answer. (there is a trade-off between increasing the transmit power (more coverage but also higher interference) and decreasing it (the converse))

香农公式  $C = B \cdot \log_2(1 + S/(I+N))$ : 增大带宽 $B$ 不一定总是好事, 因为噪声 $N=N_0B$ 也会跟着增大, 两者要权衡; 增大发射功率也不是越大越好, 因为无线信道是共享广播的, 你功率越大对别人的干扰也越大。